**Predicting Intrinsic Clearance Through Quantum-Informed Embedding of Molecular Surface in Machine Learning**

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**Abstract:** Intrinsic Clearance (CLint) measures how well a liver can metabolize a drug and is essential in early-stage drug development. By knowing a drug's CLint value, pharmacists may determine its viability, concentration, and dosage frequency. If able to predict CLint values accurately, researchers could reduce experimental costs and accelerate the identification of promising drug candidates before moving to clinical trials. While methods to encode molecular structure for Machine Learning (ML) prediction exist, they commonly focus on encoding structural information over quantum-electronic information. Structure-based encodings make approximations of what quantum electronic information directly captures along the molecular surface. Directly capturing electronic attributes is important due to their influence on intermolecular interactions, which largely determine a molecule's reactivity and metabolism. A newer approach, MEMS (Manifold Embedding of Molecular Surface), accounts for this by using quantum-informed encoding to improve predictive performance in ML models. This study evaluates whether encoding quantum-electronic surface information via MEMS can improve CLint prediction over structure-based ML approaches. Using a curated dataset of 1069 small molecules, we gathered molecular surface information via quantum mechanical methods (Density Functional Theory) and encoded it with MEMS before training a ML model to predict CLint values. We hypothesize that MEMS will outperform structure-based encodings for the ML prediction of CLint values. If successful, this would support quantum-informed encodings as a viable approach in predicting pharmacokinetic properties in small molecules and further aid researchers in the drug development process.

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